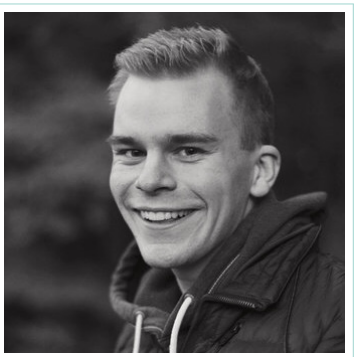


Lennart Bedarf

Personal Details

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nationality	German
date of birth	01.09.1999

Profile

PhD candidate in Biomedical Engineering at the University of Basel, working on multimodal brain tissue analysis, from temperature-based postmortem interval estimation to quantitative MRI (qMRI) and histological validation. M.Eng. in Medical Engineering (grade 1.0) with a focus on MR spectroscopy. Strong programming background (Python, MATLAB, C) and international conference experience.

University Education

06/2024 – present
PhD in Biomedical Engineering, *University of Basel*, Basel, Switzerland
Thesis topic: Multimodal Brain Tissue Analysis: From Forensic Temperature-Based Postmortem Interval Estimation to Advanced qMRI Anisotropy and Histological Validation Techniques
- Developed and evaluated forehead temperature measurements for postmortem interval estimation (first-author paper, *Int. J. Legal Med.*, 2026).
- Investigating the orientation dependency of R_2^* in human brain white matter and the effect of formalin fixation, with histological validation.

04/2022 – 01/2024
M.Eng. in Medical Engineering, *South Westphalia University of Applied Sciences (Fachhochschule Südwestfalen)*, Hagen, Germany, Overall grade: **1.0** (with honours)
Focus: advanced imaging techniques in MRI
Master's thesis (grade **1.0**): "Studies on edited magnetic resonance spectroscopy: Phantom- and *in vivo*-measurements for neuroscientific problems"

09/2018 – 04/2022
B.Eng. in Medical Engineering, *South Westphalia University of Applied Sciences (Fachhochschule Südwestfalen)*, Hagen, Germany, Overall grade: **1.5**
Bachelor's thesis (grade **1.0**): "Python-based analysis and evaluation of spectroscopic *in vivo* MR data"
German grading scale: 1.0 (best) to 5.0.

School Education

2010 – 2018
General university entrance qualification (*Abitur*), *Zeppelin-Gymnasium*, Lüdenscheid, Germany

Work Experience

12/2022 – 06/2024 **Working Student**, *TEBIT GmbH & Co. KG*, Meinerzhagen, Germany
Part-time position in high-precision medical engineering alongside studies.

2020 – 2022 **COVID-19 Tester (part-time)**, *Johanniter Seniorenheime GmbH*, Lüdenscheid, Germany
Performed rapid antigen tests for visitors of a nursing home.

Scientific Publications

Journal Articles

2026 **L. Bedarf**, K. Gerlach, H. Wittig, E. Scheurer, C. Berger, C. Lenz: "Evaluating the forehead temperature for estimating the postmortem interval". *International Journal of Legal Medicine*, 2026. DOI: 10.1007/s00414-026-03736-x

2025 J. Bottoni, H. Wittig, T. Rost, A. Schocker, P. Wild, U. Nachbur, D. Neuhaus, **L. Bedarf**, K. Gerlach, E. Scheurer, C. Lenz: "Detection of gunshot residues using infrared photography: influence of ammunition type, surface color and blood contamination". *International Journal of Legal Medicine*, 2025. DOI: 10.1007/s00414-025-03609-9

Conference Abstracts

2026 **L. Bedarf**, D. Neuhaus E. Scheurer, C. Lenz: "Incomplete reversibility of fixation-induced changes in postmortem human brain white matter MRI following rehydration". *ESMRMB Annual Meeting*, Girona, 2026.

2026 **L. Bedarf**, D. Neuhaus E. Scheurer, C. Lenz: "Fixation and rehydration influence postmortem MRI parameters". *Postmortem Research in Dementia Workshop*, New York, 2026.

2026 **L. Bedarf**, K. Gerlach, H. Wittig, E. Scheurer, C. Lenz: "Time of death estimation using forehead temperature measurements". *IAFS Triennial Meeting*, Sofia, 2026.

2025 **L. Bedarf**, D. Neuhaus, M. J. Wendebourg, R. Schlaeger, C. Birkl, E. Scheurer, C. Lenz: "Quantifying the impact of formalin fixation on R_2^* orientation dependency in human brain white matter" (poster PG253). *ESMRMB Annual Meeting*, Marseille, 2025.

2023 **L. Bedarf**, A. Holl, H. Nawrath: "Determination of *T1* and *T2* relaxation times of phosphorus-31 in a low-field MRI by developing a phosphorus-31 resonant low-field MRI coil". *Proc. International Society for Magnetic Resonance in Medicine (ISMRM)*, 2023.

Awards, Scholarships and Grants

2026 **Best Poster Prize (2nd Prize)**, *European Society for Magnetic Resonance in Medicine and Biology - 42nd Annual Scientific Meeting*, Girona, Spain
Outstanding Poster, 100€

2026 **Best Poster Prize**, *European Society for Magnetic Resonance in Medicine and Biology - Microstructural Imaging Workshop*, Girona, Spain
Outstanding Poster, 100€

2026 **Trainee Travel Award**, *Postmortem Research in Dementia Workshop*, New York, USA
Travel grant, USD 200

2026 **Travel grant (*Reisefonds*)**, *University of Basel*, Basel, Switzerland
For the IAFS Triennial Meeting in Sofia, CHF 1,101

09/2021 – 08/2023 **Deutschlandstipendium**, *Federal Ministry of Education and Research (BMBF)*, Germany
Merit-based scholarship for outstanding academic performance

Academic Service

2025 Chairperson, ESMRMB Annual Meeting, Marseille

Additional Training

2025 University of Basel: Presentation Training (02/2025); Good Scientific Practice and Its Pitfalls in Everyday Research (06/2025); Self-Branding and Self-Promotion (10/2025); Stress Management in Performance Situations (10/2025)

Skills and Interests

Languages

German Native
English Fluent
Latin Basic *Kleines Latinum*

Technical Skills

Programming Python, MATLAB, C
Engineering CAD, SolidWorks
Tools Linux, Git, LaTeX, Microsoft Office

Interests

Handball Since 2004
Cooking Catered for 30 people during a week-long youth camp
Volunteering Evangelische Versöhnungskirchengemeinde Lüdenscheid
Competition Successful participation in the international “Formula 1 in Schools” competition

Other

Driving licence Class B

Basel, 22.10.2026
Lennart Bedarf